

15	A man of mass 60 kg stands on an oscillating platform. The platform oscillates with a frequency of 0.50 Hz and an amplitude of 0.20 m. What is the minimum normal contact force exerted by the platform on the man?							
	A	zero	B	120 N	C	470 N	D	590 N

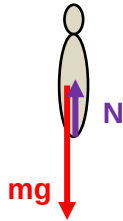
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L2 Answer: C

Let N be the Normal Contact Force.

The **minimum N** occurs at the **HIGHEST point** of the motion (i.e. UPPER amplitude position):

Draw a free body diagram of the man at the HIGHEST point:



mg downwards, greater magnitude than N upwards.

Also, at Highest point, a is maximum.

By Newton's 2nd law of motion: $mg - N = ma_{\max}$ ---(1)

From SHM equation: $a_{\max} = \omega^2 x_0$ ---(2)

Sub (2) into (1):

$$mg - N = m\omega^2 x_0$$

$$N = mg - m\omega^2 x_0 = (60)(9.81) - (60)(2\pi(0.50))^2(0.20) = 470.1647 = 470 \text{ N}$$